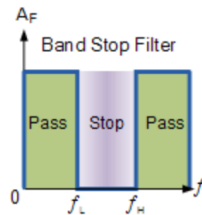
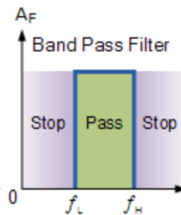
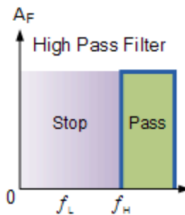
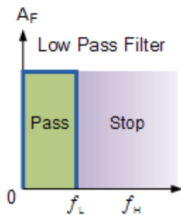
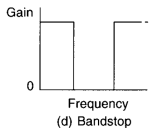
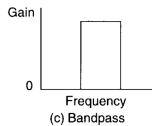
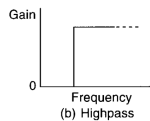
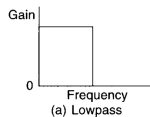
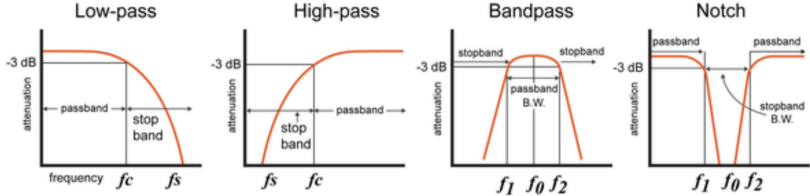
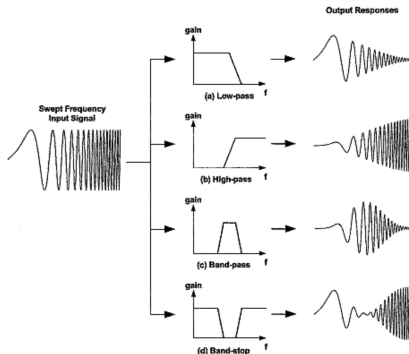


Filter design

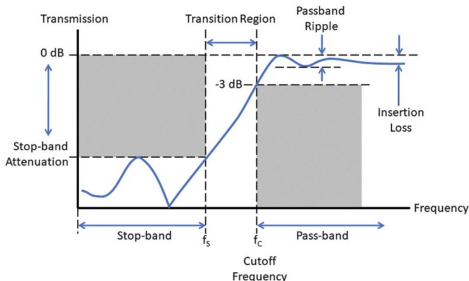




- f_c : *cutoff frequency* where attenuation begins. Typically at -3dB gain.
- f_s : *stop frequency* where stopband requirement is met. Typically at -40dB gain.



Filtering a chirp signal.



- ➊ **Passband:** is the frequency range where gain ≈ 1 .
- ➋ **Stopband:** is the frequency range where gain ≈ 0 .
- ➌ In practice passband(stopband) will deviate from the ideal value of one(zero). These **ripples** may have a specified maximum quotient between the highest and lowest gain.
- ➍ **Cutoff Frequency:** The frequency, f_c , at which the output signal has one-half the power of the input signal. This is useful in specifying where the pass-band ends and where the stop-band begins. The cutoff frequency is also referred to as the -3 dB frequency ($-3 \text{ dB} = 1/2 \rightarrow 1/\sqrt{2}$ in amplitude).
- ➎ **Bandwidth:** If a filter passes or rejects only a range of frequencies acting like both a high-pass and a low-pass filter, there will be two cutoff frequencies. The higher one is f_H and the lower is f_L .

- ➏ In practice there is a finite width **transition band** between the passband and stopband.
- ➐ **Order** of a filter: the number of poles it uses in the z-domain, and equivalently the number of delay elements necessary to implement it.
- ➑ **Ripple:** This is the variation in attenuation, specified in dB, of the input signal in the filter's pass-band or stop-band. Variation in the stop-band is also called ripple.
- ➒ **Rolloff:** The slope of the filter's response in the transition region between the pass-band and stop-band specified in dB/octave or dB/decade.
- ➓ **Insertion Loss (IL):** The minimum attenuation in the filter's pass-band.
- ➔ **Stop-band Attenuation and Notch Depth:** In the stop-band, a minimum amount of attenuation is required of the filter. If the filter rejects a narrow range of frequencies with a stop-band in between two pass-bands, notch depth is the maximum attenuation between those two pass-bands. Both of these parameters are specified in dB.
- ➕ Trade-off between small transition band, a low order, a low ripple amplitude, or even an absence of ripples.
- ➖ Design techniques for analog filters (involving capacitors, resistors, coils) can also be used to design digital IIR filters.

Recap: Filter classification

- Domain: analog/digital; time vs frequency (FFT)
- Structure: FIR / IIR
- Frequency type: low-pass, high-pass, band-pass, band-stop, all-pass
- Stability/causality: stable/unstable, causal/non-causal
- Phase: linear-phase, minimum-phase, all-pass (delay shaping)
- Design family: Butterworth, Chebyshev I/II, Elliptic, Bessel
- Realization: direct, cascade (biquad), parallel, lattice
- Adaptivity: fixed vs adaptive (LMS, RLS)

- ① **Direct design:** reasonable results for fairly simple filters
- ② **Digitizing analogue filters:** for more sophisticated filters. Classic analogue filters such as Butterworth, Chebyshev or elliptic can be discretized using s -to- z transformation methods. The “translation” is approximate, e.g., the Nyquist frequency limit.
 - ① **bilinear transform**
 - ② **impulse-invariant**
 - ③ **pole-zero matching technique**

The direct design of IIR filters

- poles and zeros are placed on the z-plane in an attempt to achieve the required frequency response.
- trial & error based
- EASY for digital filters (get the algorithm's parameters using Python/Matlab)



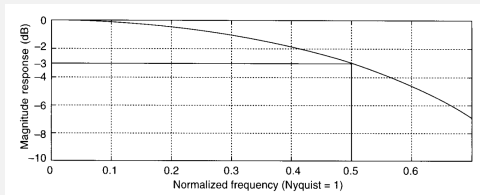
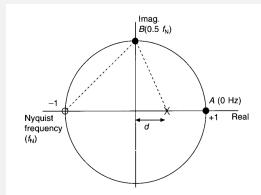
- HARD for analogue filters (to get the hardware right)



Example: lowpass filter

Requirements:

- 1 Direct current gain of 1.
- 2 Cut-off frequency of $0.25f_s$.
- 3 Transfer function of the form : $H(z) = k \frac{z+a}{z+b}$.



$$\text{Gain} = k \frac{\prod \text{zero distances}}{\prod \text{pole distances}}, \quad \text{Gain}_{\text{Nyquist}} = 0 \rightarrow a = 1.$$

$$\text{Gain}_A = k \frac{2}{1-d}, \quad \text{Gain}_B = k \frac{\sqrt{2}}{\sqrt{1+d^2}} \rightarrow \frac{\text{Gain}_B}{\text{Gain}_A} = \frac{\sqrt{2}(1-d)}{2\sqrt{1+d^2}} \stackrel{\text{req.2}}{=} \frac{1}{\sqrt{2}} \rightarrow d = 0 \rightarrow b = 0.$$

$$\text{Gain}_A = k \frac{2}{1-d} = 2k \stackrel{\text{req.1}}{=} 1 \rightarrow k = \frac{1}{2}.$$

$$H(z) = \frac{1}{2} \frac{z+1}{z} \rightarrow \left| H(e^{i\omega T}) \right| = \left| H(e^{i\pi(2f/f_s)}) \right| = \left| H(e^{i\pi x}) \right| = \frac{1}{2} \sqrt{(1 + \cos^2(\pi x)) + \sin^2(\pi x)}, \quad x \in [0, 1].$$

Example: notch filter

Requirements:

- 1 Notch frequency of 20 Hz.
- 2 Attenuation of > 40 dB.
- 3 Bandwidth < 4 Hz.
- 4 Gain in the passband is 1.
- 5 f_s is 160 Hz.

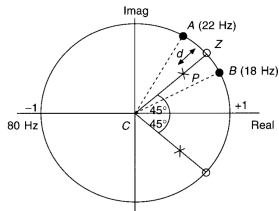
(5) $\rightarrow f_N = 80$ Hz

(1) $\rightarrow 20 \text{ Hz} \equiv 45^\circ$.

(2) $\rightarrow \text{Gain}(20 \text{ Hz}) = 0$ (zero on the unit circle at 20 Hz) $\rightarrow$ it is complex.

(3) $\rightarrow \text{Gain}(18 \text{ Hz}) = \text{Gain}(22 \text{ Hz}) = 1/\sqrt{2} \text{ Gain}(0 \text{ Hz})$

Symmetry around 20 Hz $\rightarrow$ the pole should be on the same radii as the zero
 $\rightarrow$ it is complex $\rightarrow$ we shall also have the zero's and pole's complex conjugate.

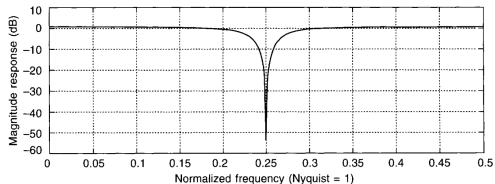


$$\angle ACZ = \frac{2}{f_N} \cdot \pi \approx 0.08 \text{ rad} \ll 1 \rightarrow \angle AZP \approx 90^\circ \rightarrow AP \approx \sqrt{d^2 + 0.08^2}$$

$$\text{Gain} = k \frac{\prod \text{zero distances}}{\prod \text{pole distances}},$$

Let $k = 1$

$$\frac{1}{\sqrt{2}} = \frac{AZ}{AP} = \frac{0.08}{\sqrt{d^2 + 0.08^2}} \rightarrow d = 0.08.$$



$$\rightarrow \text{zeros at } \cos 45^\circ \pm i \sin 45^\circ, \quad \text{poles at } 0.92(\cos 45^\circ \pm i \sin 45^\circ).$$

$$H(z) = \frac{(z - \cos 45^\circ - i \sin 45^\circ)(z - \cos 45^\circ + i \sin 45^\circ)}{[z - 0.92(\cos 45^\circ - i \sin 45^\circ)][z - 0.92(\cos 45^\circ + i \sin 45^\circ)]} = \frac{z^2 - 1.414z + 1}{z^2 - 1.302z + 0.847}.$$